Major Assignment – 3

Part A – Theory Questions

Q1.

(a) Define sample space and event. Give an example.
(b) Write the formula for joint probability of two events A and B, when they are (i) mutually exclusive, (ii) independent.

Q2.

(a) State Bayes' theorem.
(b) Explain each term of above theorem with an example.

Q3.

(a) What is a Bernoulli random variable?
(b) Why is a Bernoulli distribution considered a special case of the Binomial distribution?

Q4.

(a) Explain the difference between discrete and continuous random variables.
(b) Explain the role of covariance in measuring relationships between variables.

Q5.

(a) Define MAP (Maximum a Posteriori) hypothesis.
(b) Explain why Bayesian methods are important in machine learning.

Q6.

(a) What is a Naïve Bayes classifier?
(b) How does a Naïve Bayes classifier perform classification? Explain why Naïve Bayes works well even with limited training data.

Q7.

(a) What is a Bayesian Belief Network?
(b) Analyze the advantage of Bayesian Belief Networks over Naïve Bayes classifier.

Q8.

(a) Define classification in supervised learning.
(b) Explain the steps involved in classification learning.

Q9.

(a) Define k-Nearest Neighbours (kNN) algorithm. Explain why the choice of k is important in kNN.
(b) Why kNN is called a lazy learner? Explain the strength and weakness of kNN algorithm.

Q10.

(a) What is a decision tree?
(b) Discuss the strength and weakness of decision trees.

Q11.

(a) What is a Random Forest?
(b) Compare decision trees and Random Forests approaches.

Q12.

(a) Define Support Vector Machine (SVM) concept.
(b) Analyze how SVM handles non-linearly separable data.

Q13.

(a) Define regression in supervised learning.
(b) What are the key assumptions of linear regression?

Q14. Explain why multicollinearity is a problem in regression. What are the remedies to overcome multicollinearity.

Q15. Explain why multiple linear regression is useful in real-world problems.

Part B – Lab Questions

Q1. Write a Python program to generate Bernoulli and Binomial distributions using NumPy and calculate their mean and variance.

Q2. Write a Python program to generate random samples from a Normal distribution and plot its Probability Density Function (PDF).

Q3. Write a Python program to compute and display the covariance and correlation coefficient between two random variables.

Q4. Write a Python program to verify the Central Limit Theorem (CLT) by plotting the distribution of sample means.

Q5. Write a Python program to implement kNN classifier for a given dataset. (Iris.csv, apndcts.csv, btisuue.csv)
The program should:

- Load a dataset from a CSV file using Pandas
- Separate predictor variables and target class labels
- Split the dataset into training and testing sets
- Train a kNN model using the training data
- Predict class labels for the test data
- Compute and display the accuracy of the model